

# Hematology Lymphoid / Plasma Cell Malignancy Codex Guide

- Block: Hematology
- Method: Codex-authored manual distillation from extracted lecture markdown in this workspace.
- Sources:
  - 2026\_Yui\_Lymph\_Node\_Architecture\_Refresher.pdf
  - 2026\_Yui\_Low\_Grade\_Lymphoma-1.pdf
  - 2026\_Yui\_High\_Grade\_Lymphoma.pdf
  - 2026\_Yui\_Hodgkin\_Lymphoma.pdf
  - 20210311\_Borowitz\_Chronic\_Lymphocytic\_Leukemia.pdf
  - 20210317\_Ambinder\_HTLV.pdf
  - 2025\_Yui\_B\_Cell\_Lymphoma\_Pharmacology.pdf
  - 2025\_Imus\_Plasma\_Cell\_Disorders\_MGUS\_Multiple\_Myeloma.pdf
  - OSFM Heme Workshop Small Group 2025.pdf

## One-Minute Mental Model

Lymphoid malignancy questions are easiest when you place the malignant cell on the lymphocyte maturation map.

- **Low-grade lymphomas** are mature lymphocyte accumulation disorders: slow, indolent, often incurable but controllable, driven by survival/apoptosis escape.
- **High-grade lymphomas** are rapid proliferation disorders: fast-growing masses, B symptoms, high LDH/uric acid, tumor lysis risk, often chemo-sensitive and sometimes curable.
- **Hodgkin lymphoma** is its own pattern: rare malignant Reed-Sternberg cells in a large inflammatory background, contiguous spread, excellent prognosis.
- **Plasma cell disorders** are monoclonal immunoglobulin disorders ranging from MGUS to smoldering myeloma to symptomatic multiple myeloma with CRAB end-organ damage.

The recurring exam move is to match **clinical tempo** + **immunophenotype** + **translocation/mutation**.

## High-Yield Lecture Figures

### Lymph Node Zones

- **The Cortex:** Home to the follicles (B cells).
- **The Paracortex:** The region between follicles. This is where T-cells live and where high endothelial venules allow lymphocytes to enter from the blood.
- **The Medulla:** Contains medullary cords (plasma cells) and medullary sinuses (macrophages) leading to the hilum

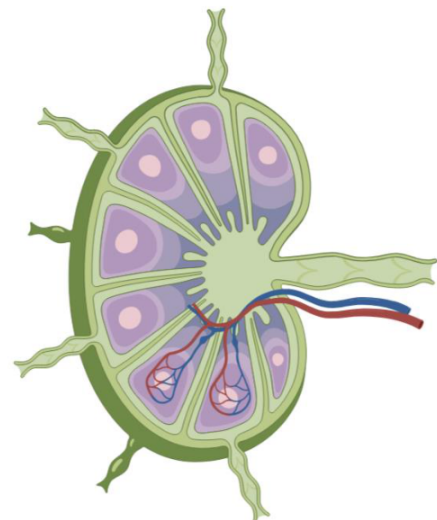
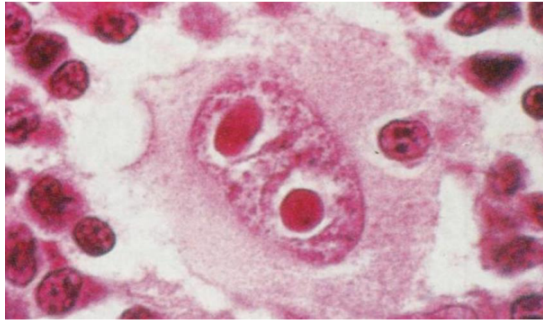


Figure 1: Lymph node zones: cortex, paracortex, and medulla.

## Reed-Sternberg Cells

- Large lymphoid cells with complex nuclear features
- Classically binucleated with prominent eosinophilic nucleoli (“**Owl eyes**”)



Dorothy Reed

Figure 2: Reed-Sternberg cells with classic bilobed nuclei and prominent nucleoli.

## Progression: MGUS → SMM → MM

- MGUS: asymptomatic clone, ~1% per year progression
- Smoldering MM: ~10% risk of progression per year for first 5 years
- Myeloma: symptomatic with end-organ damage
- **Accumulating mutations + bone marrow microenvironment changes** drive progression

|                         | MGUS | Smoldering MM | Multiple Myeloma |
|-------------------------|------|---------------|------------------|
| Monoclonal Protein      | x    | x             | x                |
| CRAB Criteria           |      |               | x                |
| Clonal PCs, Bone marrow | <10% | > 10%         | >10%             |

Figure 3: Progression from MGUS to smoldering myeloma to multiple myeloma.

## Highest-Yield Concepts

### Lymph Node Architecture and Lymphoma Diagnosis

- Lymph node cortex contains follicles/B cells; paracortex contains T cells and high endothelial venules; medulla contains plasma-cell-rich cords and macrophage-rich sinuses. [Lymph Node Architecture, slide 5]
- B-cell journey: marginal sinus entry -> mantle zone naive B-cell waiting room -> germinal center maturation center -> marginal zone exit hallway. [Lymph Node Architecture, slides 8-11]
- Germinal centers have a dark zone with proliferating centroblasts undergoing somatic hypermutation and a light zone with centrocyte selection; germinal center B cells express CD10. [Lymph Node Architecture, slide 10]
- Reactive hyperplasia is benign, polyclonal, preserves architecture, has polarized germinal centers, mixed B-cell maturation stages, and tingible body macrophages. [Lymph Node Architecture, slide 14]
- Malignant lymphoma is clonal, light-chain restricted, effaces architecture, and may have follicular pattern without normal apoptosis or diffuse sheets of identical cells. [Lymph Node Architecture, slide 15]
- Painless lymphadenopathy suggests malignancy or chronic inflammation; painful lymphadenopathy more often suggests infection. [Lymph Node Architecture, slide 13]

### Low-Grade Lymphomas

- Low-grade lymphomas are mature lymphocyte accumulation disorders driven by proliferation plus apoptosis evasion, not primarily rapid division. [Low-Grade Lymphoma, slide 5]
- Many low-grade B-cell lymphomas are driven by translocations involving the highly active IgH locus on chromosome 14, placing an oncogene under strong enhancer control. [Low-Grade Lymphoma, slide 6]
- Low-grade lymphomas usually occur in older adults, are usually B-cell, have small mature cells, low mitotic rate, indolent course, and are generally controlled rather than cured. [Low-Grade Lymphoma, slides 7-8]
- Transformation to high-grade lymphoma is signaled by new symptoms, rapidly enlarging lymphadenopathy, B symptoms, or high FDG uptake. [Low-Grade Lymphoma, slide 8; B-cell Pharmacology, slides 14-16]
- Diagnosis requires adequate tissue, often excisional biopsy, because architecture matters; diagnosis combines histology, flow cytometry, and molecular/genetic data. [Low-Grade Lymphoma, slide 9]

### Follicular Lymphoma

- Follicular lymphoma arises from germinal center B cells and is the most common low-grade lymphoma. [Low-Grade Lymphoma, slide 10]
- Presentation: adults in 5th/6th decades, painless generalized lymphadenopathy, often asymptomatic, with spleen and marrow involvement common. [Low-Grade Lymphoma, slide 11]
- Pathophysiology: t(14;18) moves BCL2 on chromosome 18 to IgH on chromosome 14, causing BCL2 overexpression and apoptosis blockade. [Low-Grade Lymphoma, slide 12]
- Histology shows follicular pattern with disrupted zones; cells are small cleaved centrocytes. Immunophenotype is CD5-, CD10+, CD20+. [Low-Grade Lymphoma, slides 13-14]
- Clinical course: watch-and-wait is often appropriate; early treatment does not improve overall survival. Therapy can include rituximab, chemotherapy, immunotherapy, or targeted therapy when symptomatic/progressive. [Low-Grade Lymphoma, slide 15]

### CLL/SLL

- CLL is the most common leukemia in the Western world, typically in older adults, median age 65-70, and more common in males. [CLL, slide 3]
- CLL has increased mature-appearing lymphocytes in blood; marrow is hypercellular with nodular or diffuse mature lymphocyte infiltrates. [CLL, slides 5-6]
- CLL is a B-cell disease with enhanced survival rather than rapid proliferation; despite B-cell proliferation, patients have B-cell dysfunction with autoimmunity and hypogammaglobulinemia. [CLL, slide 7]
- Flow cytometry is key: CLL/SLL phenotype is CD5+, CD19+, CD20+, CD23+ with light-chain restriction. [CLL, slides 8-9; Low-Grade Lymphoma, slide 17]
- Good-risk genetic lesion: del13q. Poor-risk lesions include del11q/ATM and del17p/TP53. Unmutated B-cell receptor genes predict poorer prognosis; mutated phenotype predicts better prognosis. [CLL, slides 10-13]
- Complications include cytopenias from hypersplenism, autoimmunity, or marrow replacement; infections from neutropenia/hypogammaglobulinemia; and transformation to aggressive lymphoma. [CLL, slides 14-16]

- Management goal is disease control rather than cure. Observation is common because early treatment does not improve survival. Treat for progressive marrow failure, massive hepatosplenomegaly, organ compromise, autoimmunity, or B symptoms. [CLL, slide 18]

### **Mantle Cell, Marginal Zone, Hairy Cell, and Low-Grade T-Cell Lymphoma**

- Mantle cell lymphoma is t(11;14), placing cyclin D1 under IgH control. Cyclin D1 drives G1-to-S progression in normally quiescent mantle zone cells, making mantle cell the most aggressive of the low-grade lymphomas. [Low-Grade Lymphoma, slide 19]
- Mantle cell typically presents in older men with disseminated disease involving nodes, blood, marrow, spleen, and GI tract; B symptoms/cytopenias may occur. Immunophenotype: CD5+, CD10-, CD20+, CD23-, cyclin D1+. [Low-Grade Lymphoma, slides 20 and 22]
- Marginal zone lymphoma arises from post-germinal center marginal zone memory B cells driven by chronic antigen stimulation. Subtypes: extranodal MALT, splenic, and nodal. [Low-Grade Lymphoma, slide 23]
- Marginal zone associations: H. pylori -> gastric MALT; Sjogren -> salivary gland MALT; Hashimoto -> thyroid MALT; hepatitis C -> splenic/nodal MZL. Immunophenotype: CD5-, CD10-, CD20+. Eradicate trigger when possible. [Low-Grade Lymphoma, slides 24 and 26]
- Hairy cell leukemia is mature B-cell disease in blood, marrow, and spleen, with little nodal disease. Typical patient is middle-aged/older man with cytopenias, infections/bruising, splenomegaly, and early satiety. [Low-Grade Lymphoma, slide 27]
- Classic hairy cell leukemia has BRAF V600E, marrow reticulin fibrosis causing dry tap, hairy cytoplasmic projections, and CD11c+, CD20+, CD25+, CD103+ immunophenotype. [Low-Grade Lymphoma, slides 27 and 29]
- Low-grade T-cell lymphomas are uncommon, mature post-thymic T-cell disorders often driven by chronic antigen stimulation/immune dysregulation. Mycosis fungoides is the most common: malignant CD4+ T cells infiltrate epidermis causing patches/plaques; Sezary syndrome is leukemic phase. [Low-Grade Lymphoma, slide 30]

### **High-Grade Lymphomas**

- High-grade lymphomas are rapid proliferation disorders of immature or activated lymphocytes. They present with rapidly enlarging lymphadenopathy, systemic symptoms, high LDH/uric acid, high Ki-67, and rapid progression if untreated. [High-Grade Lymphoma, slides 5-6]
- Tumor lysis syndrome risk is high because therapy kills rapidly proliferating cells, releasing potassium, phosphate, and uric acid; monitor labs and give fluids plus allopurinol/rasburicase when needed. [High-Grade Lymphoma, slide 7]
- DLBCL is the most common NHL, accounting for 30-40% of adult lymphoma; it can occur de novo or transform from low-grade B-cell lymphoma. [High-Grade Lymphoma, slide 9]
- DLBCL arises from germinal center centroblasts and often involves BCL6 translocations; transformation can add MYC, BCL2, or TP53 lesions. [High-Grade Lymphoma, slide 10]
- DLBCL presents with rapidly enlarging mass lesions, 40% extranodal involvement, B symptoms, high cytokine burden, and uncommon marrow/leukemic phase. [High-Grade Lymphoma, slide 11]
- DLBCL treatment goal is cure; standard therapy is R-CHOP, with 60-70% cure rate depending on IPI factors: age, stage, LDH, extranodal sites, performance status. [High-Grade Lymphoma, slide 12]
- Burkitt lymphoma is the fastest-growing human tumor, Ki-67 >95%, caused by MYC translocation t(8;14). Endemic African disease is EBV-associated and jaw/facial; sporadic Western disease often abdominal/GI and less often EBV-associated. [High-Grade Lymphoma, slides 14-15]
- Burkitt morphology has monomorphic medium-sized B cells, frequent mitoses/apoptotic bodies, and starry-sky macrophages. It is a medical emergency with high tumor lysis risk but is very chemotherapy responsive, especially in children. [High-Grade Lymphoma, slides 16-17]
- Lymphoblastic lymphoma is the tissue equivalent of ALL; lymphoblastic lymphoma is usually T-cell, while leukemic ALL is more often B-cell. It arises from thymic precursor T cells and often presents as mediastinal mass in male teenagers with possible SVC syndrome. [High-Grade Lymphoma, slides 18-21]
- Lymphoblastic lymphoma cells are TdT+ lymphoblasts with dispersed chromatin and frequent mitoses. [High-Grade Lymphoma, slide 20]
- Peripheral/high-grade T-cell lymphomas are CD3+ but often lose other T-cell markers such as CD4, CD7, or CD8; prognosis is generally worse than DLBCL. [High-Grade Lymphoma, slide 22]

## Hodgkin Lymphoma

- Hodgkin lymphoma is derived from crippled germinal center B cells. Reed-Sternberg cells are the neoplastic cells but make up less than 5% of the tumor mass; inflammatory cells make up most of the lesion. [Hodgkin Lymphoma, slides 6-7]
- Reed-Sternberg cells secrete cytokines such as IL-5 and TNF-alpha, producing reactive T cells, B cells, eosinophils, and plasma cells; EBV is often implicated. [Hodgkin Lymphoma, slide 7]
- HL has bimodal age distribution: 15-35 and >55; it commonly presents with painless rubbery cervical/supraclavicular lymphadenopathy and frequent mediastinal involvement. [Hodgkin Lymphoma, slides 10-11]
- HL spreads contiguously from one lymph node group to adjacent groups and typically avoids extranodal tissues, tonsils, and Waldeyer ring. [Hodgkin Lymphoma, slide 11]
- B symptoms are fever >38 C, drenching night sweats, and unintentional weight loss; stage gets A if absent, B if present. Spleen is lymphatic tissue and does not make disease stage IV. [Hodgkin Lymphoma, slide 13]
- Diagnosis requires biopsy with architecture effacement, inflammatory background, and Reed-Sternberg cells: large binucleated cells with prominent eosinophilic “owl eye” nucleoli. [Hodgkin Lymphoma, slides 17-18]
- Classical HL immunophenotype: CD15+, CD30+, CD20-, CD45-. [Hodgkin Lymphoma, slide 19]
- Classical HL subtypes: nodular sclerosis is most common and classically mediastinal; mixed cellularity is often EBV+ with eosinophils; lymphocyte rich has best prognosis; lymphocyte depleted is more aggressive and seen in older/HIV patients. [Hodgkin Lymphoma, slide 20]
- Nodular lymphocyte-predominant HL is distinct: popcorn cells, CD20+, CD15-, CD30-, usually EBV-negative, often solitary/stage II, indolent but can transform to DLBCL. [Hodgkin Lymphoma, slides 21-22]
- HL is highly chemo/radiation-sensitive and curable even in advanced disease; long-term focus is reducing late toxicity including cardiopulmonary toxicity and secondary malignancy. [Hodgkin Lymphoma, slides 23-26]

## Adult T-Cell Leukemia/Lymphoma and HTLV-1

- ATL distinctive features include cutaneous involvement, lytic bone lesions, hypercalcemia, and “flower” leukemic blasts. [HTLV, slides 1-5]
- HTLV-1 was the first human retrovirus discovered and has limited geographic distribution including Japan, parts of Africa, the Caribbean, and South America. [HTLV, slides 6 and 8]
- HTLV-1 infects and integrates into CD4 T-cell DNA; Tax activates proliferation, regulates cell cycle, promotes resistance to cell death, and causes abnormal mitosis/aneuploidy. [HTLV, slides 9-10]
- HTLV-1 transmits through breast milk, sex, and parenteral/injection exposures. It is also associated with tropical spastic paraparesis/HTLV-associated myelopathy. [HTLV, slides 12-13]
- Latency is often decades, only about 4% of infected people develop tumor, and survival after onset of cancer is typically 6-24 months. [HTLV, slides 14-15]
- Prevention includes screening pregnant women in high-incidence areas and recommending formula instead of breastfeeding. [HTLV, slide 16]
- Treatment is cytotoxic chemotherapy; antiretrovirals have no established role; allogeneic blood cell transplantation is the only established curative therapy. [HTLV, slide 17]

## B-Cell Lymphoma Pharmacology

- Low-grade B-cell lymphoma/CLL treatment often aims for disease control, not cure; observation is appropriate when asymptomatic because earlier therapy may improve event-free/progression-free survival without improving overall survival. [B-cell Pharmacology, slide 7]
- BTK is key for B-cell development/proliferation. BTK inhibitors such as ibrutinib are oral and effective in many low-grade B-cell malignancies; adverse effects include bleeding, arthralgias, atrial fibrillation, and hypertension. [B-cell Pharmacology, slide 8]
- Venetoclax is an oral BCL-2 inhibitor; BCL-2 blocks apoptosis. Adverse effects include cytopenias, infections, nausea/vomiting, and abdominal pain. [B-cell Pharmacology, slide 11]
- Anti-CD20 antibodies such as rituximab and obinutuzumab target B-cell malignancies. In low-grade lymphomas they may control disease as single agents; in high-grade lymphomas they are combined with cytotoxic therapy to improve remission/survival. [B-cell Pharmacology, slide 12]
- Monoclonal antibodies can kill through antibody-dependent cellular cytotoxicity, complement-dependent cytotoxicity, or delivery of a conjugated cytotoxic payload. [B-cell Pharmacology, slide 13]
- Richter transformation is CLL/SLL transforming into aggressive large-cell lymphoma, signaled by rapidly enlarging nodes, B symptoms, high FDG avidity, and large atypical B cells on biopsy; treatment example is R-CHOP. [B-cell Pharmacology, slides 14-16]

## Plasma Cell Disorders

- Multiple myeloma is a malignancy of long-lived plasma cells dependent on bone marrow stromal growth factors. [Plasma Cell Disorders, slide 6]
- Myeloma primary cytogenetic abnormalities include IgH translocations on chromosome 14 or hyperdiploidy; secondary progression lesions include del17p, del13, and +1q. [Plasma Cell Disorders, slide 7]
- CRAB criteria: hypercalcemia from osteolysis, renal failure from monoclonal light chains/cast nephropathy, anemia from marrow suppression, and bone disease from lytic lesions or plasmacytomas. [Plasma Cell Disorders, slides 8 and 14]
- SPEP detects and quantifies an M-spike; immunofixation identifies isotype. Common myeloma isotypes: IgG 70%, IgA 10-15%, light-chain only 10%. [Plasma Cell Disorders, slides 10-12]
- Workup: SPEP, immunofixation, quantitative immunoglobulins, serum free kappa/lambda light chains, 24-hour UPEP for Bence Jones proteinuria, CBC, CMP, cross-sectional imaging, blood smear for rouleaux, and bone marrow biopsy with cytogenetics. [Plasma Cell Disorders, slide 13]
- Suspect myeloma with macrocytic anemia, large protein gap, persistent pain/lytic lesions, proteinuria/albuminuria, renal failure, hypercalcemia, or rouleaux. [Plasma Cell Disorders, slides 14-15]
- MGUS: M-spike or elevated light chain ratio, no CRAB criteria, and if marrow is done clonal plasma cells below 10%. Progression risk is about 1% per year. [Plasma Cell Disorders, slides 16-17 and 20]
- MGUS progression risk factors include non-IgG isotype, M-protein >1.5 g/dL, and abnormal free light-chain ratio. Follow yearly with CBC, CMP, and M-spike/light chains. [Plasma Cell Disorders, slide 20]
- Smoldering myeloma has clonal plasma cells >10% and no CRAB criteria; progression risk is about 10% per year for the first 5 years. [Plasma Cell Disorders, slides 16 and 21]
- Multiple myeloma has monoclonal protein plus clonal plasma cells >10% and CRAB/end-organ damage. [Plasma Cell Disorders, slides 16 and 21]
- Solitary plasmacytoma may have yes/no monoclonal protein, no CRAB, and none or minimal marrow involvement below 10%. [Plasma Cell Disorders, slide 22]
- Multiple myeloma risk factors include age, chemical/ionizing radiation exposure, first-degree relative with MM/blood cancer, MGUS, obesity, Agent Orange, burn pits, and 9/11 first responder exposure. [Plasma Cell Disorders, slide 23]
- R-ISS staging is prognostic only and uses beta-2 microglobulin, albumin, LDH, and high-risk cytogenetics such as del17p and t(4;14). [Plasma Cell Disorders, slide 25]
- Myeloma is generally incurable and needs lifelong treatment. Transplant-eligible patients often receive triplet/quad induction -> autologous SCT -> maintenance; non-transplant patients receive triplet/quad plus maintenance. Common regimens include VRd and Dara-VRd. [Plasma Cell Disorders, slide 26]
- Myeloma therapies include IMiDs such as lenalidomide/pomalidomide, corticosteroids, proteasome inhibitors such as bortezomib/carfilzomib, anti-CD38 daratumumab, bispecific antibodies, selinexor, and CAR-T. [Plasma Cell Disorders, slides 27-32]
- Daratumumab binds CD38 on plasma cells and can cause infusion reactions, especially first IV dose. [Plasma Cell Disorders, slide 28]
- Lytic bone supportive care includes bisphosphonates such as zoledronic acid or RANKL inhibitor denosumab. [Plasma Cell Disorders, slide 29]
- Waldenstrom macroglobulinemia/IgM paraprotein can cause hyperviscosity with headaches, blurred vision, epistaxis, shortness of breath, and bleeding; urgent treatment is plasma exchange before rituximab if hyperviscosity is present. [Workshop, slide 4]

## Must-Know Comparisons

| Disease             | Cell/stage                  | Genetic/IHC anchor                 | Clinical anchor                                                 |
|---------------------|-----------------------------|------------------------------------|-----------------------------------------------------------------|
| Follicular lymphoma | Germinal center B cell      | t(14;18), BCL2, CD5-, CD10+, CD20+ | Painless generalized LAD; watch-and-wait                        |
| CLL/SLL             | Mature B cell               | CD5+, CD19+, CD20+, CD23+          | Older adult, lymphocytosis, smudge cells, hypogammaglobulinemia |
| Mantle cell         | Mantle zone B cell          | t(11;14), cyclin D1, CD5+, CD23-   | Older men, disseminated, more aggressive                        |
| Marginal zone/MALT  | Marginal zone memory B cell | CD5-, CD10-, CD20+                 | Chronic antigen stimulation; H. pylori/Sjogren/Hashimoto/HCV    |
| Hairy cell          | Post-GC memory B cell       | BRAF V600E, CD11c/CD25/CD103       | Cytopenias, splenomegaly, dry tap                               |

| Disease                  | Cell/stage                      | Genetic/IHC anchor                      | Clinical anchor                            |
|--------------------------|---------------------------------|-----------------------------------------|--------------------------------------------|
| DLBCL                    | GC centroblast/activated B cell | CD20+, often BCL6                       | Rapid mass, B symptoms, R-CHOP, curable    |
| Burkitt                  | GC B cell                       | MYC t(8;14), Ki-67 >95%                 | Fastest tumor, starry sky, TLS risk        |
| T-lymphoblastic lymphoma | Thymic precursor T cell         | TdT+                                    | Teen male, mediastinal mass, SVC syndrome  |
| Classical HL             | Crippled GC B cell              | CD15+, CD30+, CD20-, CD45-              | Contiguous spread, inflammatory background |
| NLPHL                    | GC B cell                       | CD20+, CD15-, CD30-                     | Indolent, popcorn cells, can transform     |
| Multiple myeloma         | Plasma cell                     | Monoclonal protein, marrow plasma cells | CRAB, lytic lesions, renal failure         |

## Common Confusions To Avoid

- **Reactive hyperplasia preserves architecture; lymphoma effaces it.** Clonality/light-chain restriction is a key malignant clue.
- **Low-grade does not mean harmless.** It is indolent but often incurable and can transform.
- **High-grade does not mean hopeless.** It is aggressive but often chemo-sensitive and sometimes curable.
- **Follicular lymphoma and reactive follicles both look follicular.** Follicular lymphoma lacks normal apoptosis architecture and has BCL2 upregulation.
- **CLL and mantle cell both express CD5.** CD23 favors CLL; cyclin D1 and CD23 negativity favor mantle cell.
- **Hodgkin lymphoma mass is mostly inflammatory cells, not tumor cells.** Reed-Sternberg cells are rare but diagnostic.
- **Classical HL and NLPHL have opposite immunophenotypes.** Classical is CD15/CD30; NLPHL keeps CD20.
- **Burkitt and DLBCL are both high-grade B-cell lymphomas.** Burkitt is MYC-driven, Ki-67 >95%, starry sky, emergency/TLS risk.
- **MGUS is common and not automatically cancer.** CRAB/end-organ damage separates myeloma from precursor states.
- **Light-chain myeloma may not have a large protein gap.** Use free light chains and urine studies when suspicion persists.
- **Rituximab can worsen IgM hyperviscosity initially.** Treat symptomatic hyperviscosity with plasma exchange first.

## Rapid Review

- Lymph node zones: cortex/B follicles, paracortex/T cells, medulla/plasma cells/macrophages.
- Low-grade lymphoma: mature cells, slow growth, apoptosis escape, control not cure, transformation risk.
- Follicular: t(14;18), BCL2, CD10+, watch-and-wait if asymptomatic.
- CLL: mature lymphocytosis, CD5/CD23, del17p bad, hypogammaglobulinemia/AIHA, observe unless symptomatic/progressive.
- Mantle: t(11;14), cyclin D1, CD5+, CD23-, older men, disseminated/aggressive.
- Marginal zone/MALT: chronic antigen stimulation; H. pylori gastric MALT can remit with eradication.
- Hairy cell: BRAF V600E, pancytopenia/splenomegaly, dry tap, CD11c/CD25/CD103.
- DLBCL: most common NHL, rapidly enlarging mass, B symptoms, BCL6, R-CHOP, curable.
- Burkitt: MYC t(8;14), Ki-67 >95%, starry sky, TLS risk.
- T-lymphoblastic lymphoma: teen male mediastinal mass, TdT+.
- Classical HL: Reed-Sternberg, CD15/CD30, contiguous spread, excellent prognosis.
- ATL: HTLV-1, CD4 T cells, skin/lytic bone/hypercalcemia/flower cells, allo transplant only established cure.
- Myeloma: CRAB, SPEP/IFE/FLC/UPEP, marrow biopsy, lifelong therapy.

## Active Recall Questions

1. How do reactive hyperplasia and malignant lymphoma differ architecturally and clonally?
2. What happens in the mantle zone, germinal center, and marginal zone?

3. Why are IgH translocations common in B-cell lymphomas?
4. What makes low-grade lymphomas indolent but generally incurable?
5. Which translocation drives follicular lymphoma, and what does BCL2 do?
6. How do you distinguish CLL/SLL from mantle cell lymphoma by immunophenotype?
7. What genetic findings are poor prognosis in CLL?
8. What clinical change suggests Richter transformation?
9. Which infection/inflammatory associations point to marginal zone lymphoma?
10. Why does hairy cell leukemia cause a dry tap?
11. Compare DLBCL, Burkitt, and lymphoblastic lymphoma by cell of origin, genetics, and presentation.
12. Why is tumor lysis syndrome a particular concern in high-grade lymphomas?
13. Why is Hodgkin lymphoma biologically different from NHL?
14. What is the classical Hodgkin immunophenotype?
15. What distinguishes NLPHL from classical HL?
16. How does HTLV-1 cause ATL, and how is it transmitted?
17. What workup identifies and characterizes a monoclonal plasma cell disorder?
18. Compare MGUS, smoldering myeloma, and multiple myeloma.
19. What are CRAB criteria?
20. Why is hyperviscosity from IgM paraprotein an emergency?